

Name:

1 pt

 Find f_y where

$$f(x,y) = \frac{7xy}{x^2 + 6y}$$

$f_y =$

1. **A** ☐ $-(14xy)/(6y + x^2)^2$
- B** ☐ $(7x^3)/(6y + x^2)^2$
- C** ☐ $-(7x)/(6y + x^2)^2$
- D** ☐ $(12xy + 7x^3)/(6y + x^2)^2$
- E** ☐ $(7x)/(6y + x^2)$
- F** ☐ $(7x^3)/(6y + x^2)$

1)	• B • false • true • false • false • false • false
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1 pt Find f_x where

$$f(x, y) = 5xy \ln(3xy)$$

$f_x =$

2. A ☐ $5/3$

B ☐ $5y \ln(3xy)$

C ☐ $5y \ln(3xy) + 5y$

D ☐ $5x \ln(3xy) + 5x$

E ☐ $5y$

F ☐ $5y \ln(3xy) + (5y)/3$

2)	• C • false • false • true • false • false • false
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1 pt Find the value of $f_y(6, 4)$ where

$$f(x, y) = \frac{8}{11xy}.$$

3. A ☐ -0.1680

B ☐ 0.0051

C ☐ 0.0076

D ☐ -0.0303

E ☐ -0.0051

F ☐ -0.0076

3)	• F • false • false • false • false • false • true
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1 pt Find f_x where
 $f(x, y) = 4e^{4xy(x+y)}$.

4. **A** ☐ $16(2xy + y^2)e^{4xy(x+y)}$

B ☐ $4(4 + x)e^{4xy(x+y)}$

C ☐ $4(4 + y)e^{4xy(x+y)}$

D ☐ $16(2xy + x^2)e^{4xy(x+y)}$

E ☐ $4e^{4(2xy+y^2)}$

F ☐ $4e^{4(2xy+x^2)}$

4)	• A • true • false • false • false • false • false
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1 pt Compute $f_x(2, 17)$ If $f(x, y) = \frac{(9x - 9y)^2}{\sqrt{y - 1}}$

5. **A** ☐ -607.50

B ☐ -67.50

C ☐ 2430.00

D ☐ 81.00

E ☐ 607.50

F ☐ 9.00

5)	• A • true • false • false • false • false • false
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1 pt For the function

$$z = \frac{5xy^2}{x^2y^3 + 3},$$

the first partial derivative $\frac{\partial z}{\partial y}$ is

6. $A \circ 5/(2xy)$

- $$\mathbf{B}\bigcirc (30x^3y^3 - 3x^2)/(x^2y^3 + 3)^2$$

- $$\mathbf{C}\bigcirc (5xy(x^3y^4-6xy))/(x^2y^3+3)^2$$

- $\mathbf{D} \bigcirc 10/(3xy)$

- $$\mathbf{E}\bigcirc (10xy - 3y^2)/(x^2y^3 + 3)^2$$

- $$\mathbf{F} \circ (5xy(6 - x^2y^3))/(x^2y^3 + 3)^2$$

6)	• F • false • false • false • false • false • true
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1 pt Compute $\frac{\partial z}{\partial y}$ for the following function:

$$z = ye^{4x^2-10y-5xy}$$

7. ☐ A $e^{4x^2-10y-5xy} + (-10 - 5x)e^{4x^2-10y-5xy}$
- ☐ B $e^{4x^2-10y-5xy} + ye^{4x^2-10y-5xy}$
- ☐ C $e^{4x^2-10y-5xy}$
- ☐ D $(-10 - 5x)ye^{4x^2-10y-5xy}$
- ☐ E $(-10 - 5x)e^{4x^2-10y-5xy}$
- ☐ F $e^{4x^2-10y-5xy} + (-10 - 5x)ye^{4x^2-10y-5xy}$

7)	• F • false • false • false • false • false • true
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1 pt The height of a hill is defined by the function $h(x, y)$ where x is the miles east and y is the miles north of the center of a city. Given

$$h(x, y) = 20(31 - 7x^2 - 3y^2 + 35xy + 45x - 20y)$$

Suppose you are standing on the hill 5 mile east and 3 miles north of the city center. What is the rate of change in the northern direction?

8. ☐ A 2740
- ☐ B 1600
- ☐ C -560
- ☐ D 137
- ☐ E 700
- ☐ F 3360

8)	• A • true • false • false • false • false • false
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1 pt The post office will accept packages whose combined length and girth is at most 41 inches. (The girth is the perimeter/distance around the package perpendicular to the length; for a rectangular box, the length is the largest of the three dimensions.) What is the largest volume in cubic inches that can be sent in a rectangular box?

9. ☐ A 420.25
☐ B 319.08
☐ C 2552.63
☐ D 1276.31
☐ E 13.67
☐ F 638.16

9)	• F • false • false • false • false • false • true
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1 pt Let the function $g(x, y) = -\frac{1}{2}x^2 + 4xy^2 - 16y^4 - 3y^2 + 10$, whose first order partial derivatives are $g_x = -x + 4y^2$ and $g_y = 8xy - 64y^3 - 6y$. Determine the number of critical points and classify them.

10. ☐ A 1 relative minima, 1 relative maxima, 1 saddle point(s).
☐ B 0 relative minima, 1 relative maxima, 1 saddle point(s).
☐ C 0 relative minima, 2 relative maxima, 0 saddle point(s).
☐ D 1 relative minima, 1 relative maxima, 0 saddle point(s).
☐ E 1 relative minima, 2 relative maxima, 0 saddle point(s).
☐ F 0 relative minima, 1 relative maxima, 0 saddle point(s).

10)	• F • false • false • false • false • false • true
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1 pt Consider the function $f(x, y) = 14 - 3y - 6x^2y - 5x^2 - \frac{43}{10}y^2$.

How many relative maxima, relative minima, and saddle points does $f(x, y)$ have?

11. ☐ A 0 relative minima, 1 relative maxima, 1 saddle point(s).
☐ B 0 relative minima, 2 relative maxima, 2 saddle point(s).
☐ C 1 relative minima, 2 relative maxima, 2 saddle point(s).
☐ D 1 relative minima, 1 relative maxima, 1 saddle point(s).
☐ E 0 relative minima, 1 relative maxima, 2 saddle point(s).
☐ F 1 relative minima, 1 relative maxima, 2 saddle point(s).

11)	• E • false • false • false • false • true • false
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1 pt Find and classify the critical points of $f(x, y) = -7(x + 1)^2 + 12y^3 - 25y + 11$.

12. ☐ A There is one relative minimum and one saddle point.
- ☐ B There is one relative maximum and one relative minimum.
- ☐ C There are two saddle points.
- ☐ D There is one relative maximum and one saddle point.
- ☐ E There are two relative maxima.
- ☐ F There are two relative minima.

12)	• D • false • false • false • true • false • false
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1 pt Find the value of x that (locally) maximizes the function $f(x, y) = 24x + \frac{120}{7}y - 21x^2 - 6xy - 3y^2 - 1$. If $f(x, y)$ does not have a local maximum, choose "None".

13. ☐ A -0.81
- ☐ B -1.09
- ☐ C 0.19
- ☐ D -1.14
- ☐ E 0.18
- ☐ F "None"

13)	• C • false • false • true • false • false • false
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1 pt Find the number relative maxima, relative minima, and saddle points of $f(x, y)$ if they exist. $f(x, y) = -3x^3 + \frac{1}{2}y^2 + 6xy - y + 2$.

14. ☐ A 2 relative minima, 1 relative maxima, 1 saddle point(s).
☐ B 1 relative minima, 1 relative maxima, 1 saddle point(s).
☐ C 1 relative minima, 0 relative maxima, 1 saddle point(s).
☐ D 1 relative minima, 0 relative maxima, 2 saddle point(s).
☐ E 2 relative minima, 0 relative maxima, 1 saddle point(s).
☐ F 2 relative minima, 0 relative maxima, 2 saddle point(s).

14)	• C • false • false • true • false • false • false
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1 pt Find the number relative maxima, relative minima, and saddle points of $f(x, y)$ if they exist. $f(x, y) = \frac{1}{3}x^3 - 3xy - 2y^2 - 10x + 4$.

15. ☐ A 0 relative minima, 2 relative maxima, 1 saddle point(s).
☐ B 1 relative minima, 2 relative maxima, 1 saddle point(s).
☐ C 0 relative minima, 1 relative maxima, 2 saddle point(s).
☐ D 1 relative minima, 1 relative maxima, 1 saddle point(s).
☐ E 1 relative minima, 1 relative maxima, 2 saddle point(s).
☐ F 0 relative minima, 1 relative maxima, 1 saddle point(s).

15)	• F • false • false • false • false • false • true
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1 pt Find the number relative maxima, relative minima, and saddle points of $f(x, y)$ if they exist. $f(x, y) = \frac{5}{3}x^3 + 2xy - 2y^2 - 6x + 2$.

16. ☐ A 0 relative minima, 2 relative maxima, 1 saddle point(s).
☐ B 0 relative minima, 1 relative maxima, 1 saddle point(s).
☐ C 1 relative minima, 1 relative maxima, 1 saddle point(s).
☐ D 1 relative minima, 1 relative maxima, 2 saddle point(s).
☐ E 1 relative minima, 2 relative maxima, 1 saddle point(s).
☐ F 0 relative minima, 1 relative maxima, 2 saddle point(s).

16)	• B • false • true • false • false • false • false
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1 pt Find and classify the critical points of the function

$$g(u, v) = 6u^2v + 48uv + 2v^2$$

17. **A** $\bigcirc$ $(-8, 24)$ is a saddle point; $(-4, 0)$ is a relative minimum; $(0, 0)$ is a saddle point.
B $\bigcirc$ $(0, 24)$ is a saddle point; $(-4, 0)$ is a relative minimum; $(-8, 0)$ is a saddle point.
C $\bigcirc$ $(0, 24)$ is a saddle point; $(-4, 0)$ is a saddle point; $(-8, 0)$ is a saddle point.
D $\bigcirc$ $(-4, 24)$ is a saddle point; $(0, 0)$ is a relative minimum; $(-8, 0)$ is a saddle point.
E $\bigcirc$ $(0, 0)$ is a relative minimum; $(-4, 24)$ is a relative minimum; $(-8, 0)$ is a saddle point.
F $\bigcirc$ $(0, 0)$ is a saddle point; $(-4, 24)$ is a relative minimum; $(-8, 0)$ is a saddle point.

17)	• F • false • false • false • false • false • true
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1 pt A shop sells two competing brands of socks, Levis and Gap. Each pair of socks is obtained at a cost of 4 dollars per pair. The manager estimates that if he sells the Levis socks for x dollars per pair and the Gap socks for y dollars per pair, then consumers will buy $17 - \frac{5}{2}x + 2y$ pairs of Levis socks and $1 + 3x - 3y$ pairs of Gap socks. How should the manager set the prices so that the profit will be maximized? Round your answers to the nearest cent.

$(x, y) =$

18. A ☐ (21.4, 18)
B ☐ (1, 1)
C ☐ (3.67, 4)
D ☐ (4.75, 4.75)
E ☐ (4, 3.5)
F ☐ (23, 20)

18)	• F • false • false • false • false • false • true
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1 pt A manufacturer is planning to sell a new product at the price of 400 dollars per unit and estimates that if x thousand dollars is spent on development and y thousand dollars is spent on promotion, consumers will buy approximately $\frac{240y}{y+5} + \frac{200x}{x+8}$ units of the product. If manufacturing costs for the product are 110 dollars per unit, how much should the manufacturer spend on development to generate the largest possible profit?

19. A ☐ \$673
B ☐ \$13541
C ☐ \$13655
D ☐ \$585
E ☐ \$125071
F ☐ \$60209

19)	• B • false • true • false • false • false • false
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1 pt

A manufacturer makes special ornate candles at two different locations. The cost of producing x candles at location 1 is $C_1 = 0.03x^2 + 18.3x + 480$ dollars and the cost of producing y candles at location 2 is $C_2 = 0.06y^2 + 15.6y + 225$ dollars.

The candles sell for \$30 each. Assuming all candles produced will be sold, determine how many candles should be made at location 2 so that the profit will be maximized.

20. A ☐ 380
B ☐ 305
C ☐ 130
D ☐ 120
E ☐ 805
F ☐ 195

20)	• D • false • false • false • true • false • false
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